# STRUCTURAL DEFLECTION CALCULARION FOR CHEQUERD PLATE

## 1. Given Data

Plate Material: Mild Steel

Thickness (t): 8 mm

Width (b): 375 mm

Plate Length: 2540 mm

Support System:

Number of C-Channels: 5

Spacing between channels (L): 350 mm

End Support Boxes: 150 mm width

Load:

Total Load = 1 ton = 1000 kg

Force (P) = 1000 × 9.81 = 9810 N

Modulus of Elasticity for Mild Steel (E):

$E = 2 \times 10^5 \text{ N/mm}^2$

## 2. Load per Unit Length

Load assumed uniformly distributed across plate length.

w = P / Plate Length

w = 9810 / 2540

w = 3.86 N/mm

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3. Moment of Inertia of Plate Section

For rectangular plate section:

$I = (b \times t^3) / 12$

$I = (375 \times 8^3) / 12$

$I = (375 \times 512) / 12$

$\underline{I = 16000\ mm^4}$

## 4. Deflection Formula (Simply Supported Beam with UDL)

Maximum deflection:

$\delta max = (5\ w\ L^4) / (384\ E\ I)$ - Formula

## 5. Substitution of Values

$\delta = (5 \times 3.86 \times 350^4) / (384 \times 2\times10\text{^}5 \times 16000)$

Calculate $L^4$

$350^4 = 1.50 \times 10\text{^}10$

Numerator:

$5 \times 3.86 \times 1.50 \times 10\text{^}10$

$= 2.895 \times 10\text{^}11$

Denominator:

$384 \times 200000 \times 16000$

$= 1.2288 \times 10\text{^}12$

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## 6. Maximum Deflection

$\delta = (2.895 \times 10^{11}) / (1.2288 \times 10^{12})$

$\delta \approx 0.236$ mm

## 7. Allowable Deflection Check

Allowable deflection = L / 360

= 350 / 360

= 0.97 mm

## 8. Comparison

Calculated deflection = 0.236 mm

Allowable deflection = 0.97 mm

Since 0.236 << 0.97

The deflection is well within allowable limits.

## 9. Bending Stress Check

Maximum moment for UDL:

$M = wL^2 / 8$

$M = (3.86 \times 350^2) / 8$

$M = 59000$ Nmm

Section modulus:

$Z = (b \times t^2) / 6$

$Z = (375 \times 8^2) / 6$

$Z = 4000$ mm$^3$

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Stress:

$\sigma = M / Z$

$\sigma = 59000 / 4000$

$\underline{\sigma = 14.8 \text{ MPa}}$

## 10. Strength Check

Typical allowable stress for mild steel ≈ 150 MPa

Calculated stress = 14.8 MPa

Since 14.8 << 150

The plate is structurally safe.

## 11. Final Result

Plate: 8 mm Mild Steel Chequered Plate

Support spacing: 350 mm

Load: 1 ton

Maximum Deflection: 0.236 mm

Maximum Bending Stress: 14.8 MPa

## 12. Conclusion:

The Design is safe in Deflection.

The Design is safe in Bending Stress.

Under the Load Condition : 1000kg.

Deflection and stress are far below allowable limits.

The Design is Safe.

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